

Quantitative Political Analysis

Sean J. Westwood

Location: Silsby 126
Class Meetings: Wednesdays 6:30–9:30 PM (plus Sat 6/27, Tue 6/30 & Tue 8/11); June 27 – Aug 26, 2026
Office Hours: Wednesday 4:30–6:00 PM EST (before class) by appointment
Appointments: <https://calendar.app.google/DHS6U5GH5DX9wWrh6>

Course Overview

This course introduces practical uses of AI in research. You will learn to interpret published work, generate clear summaries, and support your own scholarship. The goal is to provide a shared foundation of AI skills that helps level differences in prior experience.

The growing amount of available quantitative data is changing the way we understand and engage in politics, how we implement policy, and how organizations across the world make decisions. In this course, we will learn the fundamental tools of quantitative analysis and apply them to a wide range of political and policy oriented questions. How do we predict presidential elections? Did economic voters bring Hitler to power? How does the ideology of the Supreme Court change over time? These are just a few of the questions we will work on in the course.

Students are not expected to have any programming, advanced math or statistical knowledge. The course will be centered around assignments that will help build skills from scratch. Students will leave the course equipped for work in any setting that requires a social scientific approach to data science, from policy non-profits to government, from Silicon Valley to Wall Street and beyond. There are no formal prerequisites for this course.

Teaching in the Age of AI

AI has fundamentally changed the world, and we need to work WITH it, not against it. Some professors are pretending AI doesn't exist, others are dejected by its rise, but our job is to teach students the skills they actually need. AI is what you will use in the real world—that is what this class embraces and teaches.

The traditional approach to teaching quantitative methods—memorizing syntax, debugging code line by line, struggling with programming details—is becoming obsolete. In 2025, successful data analysts don't spend their time remembering whether it's `filter()` or `subset()`. Instead,

they focus on the intellectual work: understanding what questions to ask, designing appropriate analyses, interpreting results correctly, and communicating findings effectively.

Rather than memorizing programming syntax, we focus on:

- **Understanding concepts** and their applications to real political problems
- **Interpreting results** and their political implications in context
- **Asking the right questions** and designing appropriate research strategies
- **Working effectively with AI** to accelerate analysis and overcome technical barriers
- **Verifying logic** and critically evaluating AI suggestions for accuracy
- **Recognizing limitations** of both data and analytical methods
- **Communicating findings** clearly to diverse audiences

Assessment

Quizzes (9 total, 30 min each)	60%
Final Exam	30%
Assignments and Participation	10%

Exams and Quizzes

All exams and quizzes are **closed book** and conducted **on paper**. Quizzes are 30 minutes each, administered at the **end of class** on the nine dates marked in the schedule below (9 quizzes total). Each quiz covers the material taught since the previous quiz.

Class assignments: On most class meetings, you will be given an assignment to work through. The content will be a mixture of multiple-choice, fill-in-the-blank, and matching questions, along with tasks that require reasoning and interpretation. You will have access to a computer for these assignments. You will not submit them for credit, but it will not be possible to pass quizzes and exams without completing them.

Class Format & Required Tools

Class will be a mixture of lecture and hands-on exercises.

- **During lecture portions:** Class will be closed computer/no devices to maintain focus
- **During hands-on portions:** Laptops required for coding exercises and AI-assisted analysis

Required Tools: **R** (statistical programming language), **Positron** (modern IDE optimized for data science), **Claude Plus subscription** (AI assistant for coding and analysis)

Book: There is no book. I will provide all needed materials.

Weekly Schedule

Week 1, Class 1 (Sat 6/27): Foundations

Introduction to Quantitative Political Analysis

Special Meeting · Sat 4:00–5:05 PM

- What quantitative political analysis is and why it matters
- Data analysis applications to real political problems
- Role of AI in modern political analysis

Week 2, Class 1 (Tue 6/30): Introduction to R and Data Frames

Introduction to R and Data Frames

X-Hour · 7:30–8:20 PM

- Basic R operations with variables and vectors
- Examining data frames
- Effective AI assistance for R programming

R Functions: `library(tidyverse), read_csv(), glimpse(), head(), mean(), table(), summary()`

Week 2, Class 2 (Wed 7/1): Working with Real Data & Summary Statistics

Segment A: Working with Real Data

Quiz 1

- Exploring real political datasets
- Data manipulation with tidyverse
- Handling missing values (NA) properly
- Correlation vs. causation

R Functions: `filter(), select(), arrange(), names(), is.na(), !is.na()`

Segment B: Summary Statistics

- Central tendency: mean, median, mode
- Measures of spread: range, variance, standard deviation
- When each measure is appropriate

R Functions: `group_by()`, `summarise()`, `median()`, `sd()`, `n()`, `ggplot()`, `geom_histogram()`, `geom_density()`, `geom_vline()`

Week 3, Class 1 (Wed 7/8): Research Design

Research Design

Guest Lecture

Quiz 2

- Experimental vs. observational designs
- Natural experiments
- Cross-sectional vs. longitudinal data
- Design tradeoffs and internal validity

R Functions: `count()`

Week 4, Class 1 (Wed 7/15): Transforming Variables & Causality

Segment A: Transforming and Creating Variables

Quiz 3

- Creating and modifying variables with `mutate()`
- Conditional logic and recoding data
- Logical operators and data transformation

R Functions: `mutate()`, `case_when()`, `if_else()`, `pull()`, `n_distinct()`, `across()`, `recode()`

Segment B: Causality

- Three requirements for causation: association, temporal ordering, no confounding
- Spurious correlations and confounding variables
- Using AI to identify alternative explanations

Week 5, Class 1 (Wed 7/22): Modern Causal Inference & Linear Regression

Segment A: Modern Causal Inference

Quiz 4

- Fundamental problem of causal inference
- Average Treatment Effect (ATE) calculation and interpretation
- Randomized experiments and difference-in-differences design

Segment B: Linear Regression

- Linear relationships and OLS intuition
- Model fitting and interpretation
- Political applications of regression

R Functions: `lm()`, `broom::tidy()`

Week 6, Class 1 (Wed 7/29): Plotting Data

Plotting Data

Guest Lecture

Quiz 5

- Grammar of Graphics principles
- Line plots for time series data
- Scatter plots for relationships
- Box plots for group comparisons

R Functions: `scales::percent_format()`, `geom_line()`, `geom_point()`, `geom_smooth()`, `geom_boxplot()`

Week 7, Class 1 (Wed 8/5): Multivariate Models & Categorical Predictors and Interactions

Segment A: Multivariate Models

Quiz 6

- Controlling for confounding variables
- Partial relationships and model building
- Multiple predictors

Segment B: Categorical Predictors and Interactions

- Dummy variables and reference categories
- Interpreting categorical predictors in regression
- Interaction terms and conditional relationships
- When and how to use interaction terms

R Functions: `factor()`, `levels()`

Week 8, Class 1 (Tue 8/11): Experimental Design

Experimental Design in Political Science

X-Hour · 7:30–8:20 PM

- Types of experiments: vignette, conjoint, and economic games
- Within-subjects vs. between-subjects designs
- Typical 2×2 factorial designs
- Control group strategies and placebo effects
- Design considerations: survey length and respondent fatigue
- Random assignment and treatment implementation

Week 8, Class 2 (Wed 8/12): Sampling and Uncertainty & Statistical Significance

Segment A: Sampling and Uncertainty

Quiz 7

- Understanding populations vs. samples
- Why sampling creates uncertainty
- How sample size affects confidence
- Margin of error and its calculation
- A first look at the t-distribution and critical values

Segment B: What is Statistical Significance?

- Standard deviation vs. standard error
- Variability in samples vs. variability in estimates
- Systematic method to distinguish real effects from noise
- Political examples: polls, gender gaps, campaign effects

Week 9, Class 1 (Wed 8/19): Statistical Tests & Confidence Intervals

Segment A: Statistical Tests

Quiz 8

- Putting the t-distribution to work
- One-sample t-tests for single means
- Two-sample t-tests for comparing groups
- Regression t-tests for coefficients
- Common pitfalls in hypothesis testing

R Functions: `tibble()`, `rnorm()`, `sample()`, `replicate()`

Segment B: Confidence Intervals

- From t-statistics to confidence intervals
- Interpreting confidence intervals correctly
- Precision: narrow vs. wide intervals
- Communicating effect size AND uncertainty

R Functions: `t.test()`, `confint()`, `broom::glance()`

Week 10, Class 1 (Wed 8/26): Final Preparation & Review

Final Preparation and Review

Quiz 9

- Comprehensive review of course concepts
- Practice problems integrating multiple methods
- Q&A and final exam preparation

Final Exam

Saturday, August 29, 2026 @ 8:00 AM

Mandatory Disclosures

Dartmouth's Academic Honor Principle

All students are expected to abide by Dartmouth's Academic Honor Principle. For detailed information, please visit: dartmouth.edu/judicialaffairs/honor

Accessibility Needs

Students with disabilities who may need disability-related academic adjustments and services for this course are encouraged to see me privately as early in the term as possible. Students requiring disability-related academic adjustments and services must consult the Student Accessibility Services office (Carson Hall, Suite 125, 646-9900).

Once SAS has authorized services, students must show the originally signed SAS Services and Consent Form and/or a letter on SAS letterhead to their professor. As a first step, if students have questions about whether they qualify to receive academic adjustments and services, they should contact the SAS office. All inquiries and discussions will remain confidential.

For detailed information regarding expectations of the faculty to ensure accessibility for all students, please visit: [Dartmouth's Student Accessibility Services](#)